

Manually Preparing Input Files for Surface Reactions in OpenFOAM-8

*surfaceChemistryFoam: An OpenFOAM-based library for detailed
surface chemistry in reacting flow simulations*

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Note: The preparation of input files for simulations of catalytic processes involving microkinetic surface reaction mechanisms is a critical step prior to using *catalystFoam* solver. Detailed procedure, along with illustrative examples, are provided in this document.

Prerequisite: A set of input files in CHEMKIN format is required to describe both gas-phase kinetics and microkinetic surface reactions. These files include:

- *chem.inp* (gas-phase reaction kinetics)
- *thermo.dat* (thermodynamic property data)
- *trans.dat* (transport property data)
- *surfaceChem.inp* (microkinetic mechanism for surface reactions)

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1. Introduction

To simulate reacting flow problems that include surface reactions (i.e., catalytic processes) using *catalystFoam*, parameters for evaluating reaction rates (both gas-phase and surface reactions), as well as thermodynamic and transport properties, are required. These parameters are typically provided in CHEMKIN format [1], as specified in the files *chem.inp*, *surfaceChem.inp*, *thermo.dat*, and *trans.dat*.

However, the CHEMKIN format is not directly supported by OpenFOAM (OF) input files. For gas-phase chemistry in OF, the required input files for reacting solvers (e.g., *reactingFoam*) are usually generated by converting CHEMKIN files into OpenFOAM format using the standard *chemkinToFoam* utility. In contrast, for surface reactions, the conversion process must currently be performed manually.

Therefore, it is important for users to understand the rules and conventions underlying the conversion from CHEMKIN format to OpenFOAM format, as this knowledge enables the manual creation of the necessary input files for *catalystFoam*.

It should be noted that this document focuses exclusively on the conversion process itself and does not provide detailed explanations of the CHEMKIN file format, as such information is already well documented elsewhere.

2. Gas-phase reactions

2.1. Basic rate expression

A gas-phase kinetic mechanism may include I reactions that involve K chemical species in the system. Reactions can be represented in a general form:

$$\sum_{k=1}^K v'_{ki} \chi_k = \sum_{k=1}^K v''_{ki} \chi_k \quad (i = 1, \dots, I) \quad (1)$$

where χ_k is the chemical symbol for k -th species. The stoichiometric coefficients for elementary reactions, v_{ki} , are integers. K is the total number of species.

The net production rate $\dot{\omega}_k$ is in units of $\text{mol}/\text{cm}^3/\text{sec}$ for each of the K species:

$$\dot{\omega}_k = \sum_{i=1}^I v_{ki} q_i \quad (2)$$

where the net stoichiometric coefficient, v_{ki} , is given by (the stoichiometric coefficients may be non-integers for global rate expressions):

$$v_{ki} = v_{ki}'' - v_{ki}' \quad (3)$$

The rate-of-progress variable, q_i , for the i -th reaction is given by the difference of the forward rates and the reverse rates. By default, the law of mass action is used to determine the forward and reverse rates based on the stoichiometric coefficients and the species concentrations.

$$q_i = k_{fi} \prod_{k=1}^K [X_k]^{v_{ki}'} - k_{ri} \prod_{k=1}^K [X_k]^{v_{ki}''} \quad (4)$$

where $[X_k]$ is the concentrations of species k -th, in mol/cm^3 .

The forward rate constants, k_{fi} , for the I reactions are (by default) assumed to have the Arrhenius temperature dependence given as:

$$k_{fi} = A_i T^{\beta_i} \exp\left(-\frac{E_i}{RT}\right) \quad (5)$$

where the pre-exponential factor A_i is in cgs, the temperature exponent β_i is unitless, the activation energy E_i has a unit compatible with RT , and R is the gas constant.

2.2. CHEMKIN format for gas-phase reactions

These three parameters are required input for each reaction, and they are specified in the *chem.inp* file. For example, in the *chem.inp* of the Li mechanism file (for H_2 combustion) you will see:

```

ELEMENTS
H O N
END
SPECIES
H2 O2 O OH H2O H H02 H2O2 N2
END
REACTIONS // default: A in cgs, beta unitless, Ei in cal/mol
//
H+O2=O+OH          3.547e+15      -0.406      1.6599E+4
O+H2=H+OH          0.508E+05       2.67       0.629E+04
...
END
```

2.3. OpenFOAM format for gas-phase reactions

The above mechanism in OF format is as:

```
reactions
{
  un-named-reaction-0
  {
    type          reversibleArrheniusReaction;
    reaction      "H + O2 = O + OH";
    A             3.547e+12;
    beta          -0.406;
    Ta            8352.471373; //Ta = Ei/R
  }
  un-named-reaction-1
  {
    type          reversibleArrheniusReaction;
    reaction      "O + H2 = H + OH";
    A             50.8;
    beta          2.67;
    Ta            3165.072892; //Ta = Ei/R
  }
  ...
}
```

It is of importance to note that OF using SI unit while in CHEMKIN using cgs unit.

2.4. Converting units from cgs to SI

In the cgs system the rate-of-progress always has the unit of $\text{mol}/\text{cm}^3\text{s}$, and the concentration $[X_k]$ has the unit of mol/cm^3 . Then, we can balance the unit of left-hand-side and right-hand-side (only consider the first term) of the equation 4 as:

$$\frac{\text{mol}}{\text{cm}^3\text{s}} = \frac{1}{s} \left[\frac{\text{mol}}{\text{cm}^3} \right]^{1-\sum_i^K v'_{ki}} \left[\frac{\text{mol}}{\text{cm}^3} \right]^{\sum_i^K v'_{ki}} \quad (6)$$

where K is the number of reactants in the left-hand-side of the reaction.

In the SI system the rate-of-progress always has the unit of $\text{kmol}/\text{m}^3\text{s}$, and the concentration $[X_k]$ has the unit of kmol/m^3 . Then, we can balance the unit of left-hand-side and right-hand-side (only consider the first term) of the equation 4 as:

$$\frac{\text{kmol}}{\text{m}^3\text{s}} = \frac{1}{s} \left[\frac{\text{kmol}}{\text{m}^3} \right]^{1-\sum_i^K v'_{ki}} \left[\frac{\text{kmol}}{\text{m}^3} \right]^{\sum_i^K v'_{ki}} \quad (7)$$

Now, rewriting the unit of rate constant (i.e., the first term on the right-hand-side of the equation 7) in terms of cgs units yields:

$$\frac{1}{s} \left[\frac{\text{kmol}}{\text{m}^3} \right]^{1-\sum_i^K v'_{ki}} = \frac{1}{s} \left[\frac{\text{mol } 10^3}{\text{cm}^3 10^6} \right]^{1-\sum_i^K v'_{ki}} \quad (8)$$

or

$$\frac{1}{s} \left[\frac{kmol}{m^3} \right]^{1-\sum_i^K v'_{ki}} = \frac{1}{s} \left[\frac{mol}{cm^3} \right]^{1-\sum_i^K v'_{ki}} [10^{-3}]^{1-\sum_i^K v'_{ki}} \quad (9)$$

In other words, to convert from cgs unit to SI unit the rate constant in cgs system need to be multiplied by a coefficient, *coeff*, as:

$$coeff = \frac{1}{[10^{-3}]^{(1-\sum_i^K v'_{ki})}} = [10^{-3}]^{(\sum_i^K v'_{ki}-1)} \quad (10)$$

Finally, the unit of the pre-exponential factor A_i in the rate constant must be multiplied by *coeff* when converting units from cgs to SI.

For activation energy E_i the conversion coefficient depends on the keyword of reaction line in the CHEMKIN file.

If there is no keyword (by default), E_i is in *cal/mol*. Then T_a in OF is calculated as:

$$T_a = \frac{E_i}{R_c} \quad (11)$$

where $R_c = 1.987316 \text{ cal/g.mol.K}$

If there is *JOULE/MOLE*, E_i is in *J/mol*. Then T_a in OF is calculated as:

$$T_a = \frac{E_i}{R} \quad (12)$$

where $R = 8.31451 \text{ J/kg.mol.K}$

2.5. Examples

3. Surface reactions

3.1. Basic rate expression

A micro kinetic mechanism may include I reactions that involve up to K chemical species in the system. Reactions can be represented in a general form:

$$\sum_{k=1}^K v'_{ki} \chi_k = \sum_{k=1}^K v''_{ki} \chi_k \quad (i = 1, \dots, I) \quad (13)$$

where K is the total number of species (gas, surface, bulk).

The net production rate $\dot{\omega}_k$ is in units of $\text{mol}/\text{cm}^2/\text{sec}$ for each of the K species, regardless of phase.

$$\dot{\omega}_k = \sum_{i=1}^I v_{ki} q_i \quad (14)$$

where the net stoichiometric coefficient, v_{ki} , is given by (the stoichiometric coefficients may be non-integers for global rate expressions):

$$v_{ki} = v_{ki}'' - v_{ki}' \quad (15)$$

The rate-of-progress variable, q_i , for the i -th reaction is given by the difference of the forward rates and the reverse rates. By default, the law of mass action is used to determine the forward and reverse rates based on the stoichiometric coefficients and the species concentrations.

$$q_i = k_{fi} \prod_{k=1}^K [X_k]^{v_{ki}'} - k_{ri} \prod_{k=1}^K [X_k]^{v_{ki}''} \quad (16)$$

The form of the concentrations $[X_k]$ depends upon whether species k is in the gas-phase, on the surface, or in the bulk.

The forward rate constants, k_{fi} , for the I reactions are (by default) assumed to have the Arrhenius temperature dependence given as:

$$k_{fi} = A_i T^{\beta_i} \exp\left(-\frac{E_i}{RT}\right) \quad (17)$$

where the pre-exponential factor A_i is in cgs, the temperature exponent β_i is unitless, the activation energy E_i has an unit compatible with RT , and R is the gas constant.

3.2. CHEMKIN format for surface reactions

These three parameters are required input for each reaction, and they are specified in the chem.inp file. For example, in the chem.inp of the GRI.3.0 mechanism file you will see:

```
SITE/Ru_surface/      SDEN/7.3E-7/
  Ru(s)
  NH3(s)
  N(s)
  H(s)
END
REACTIONS      MWOFF      KJ/ MOLES
NH3 + Ru(s) => NH3(s)      1.1479E+10      0.000      0.0000
```

NH3(s) + 3Ru(s) => N(s) + 3H(s)	4.7625E+37	0.000	105.80
2N(s) => N2 + 2Ru(s)	1.1802E+17	0.000	123.50
2H(s) => H2 + 2Ru(s)	1.3536E+14	0.000	67.800
NH3(s) => Ru(s) + NH3	3.2890E+19	0.000	64.000
H2 + 2Ru(s) => 2H(s)	8.9642E+12	0.000	0.0000
END			

3.3. OpenFOAM format for surface reactions

```

reactions
{
    un-named-reaction-0
    {
        type                irreversibleSurfaceArrheniusSurface;
        reaction              "NH3 + Ru(s) = NH3(s)";
        A                    1.1479e+7;
        beta                  0;
        Ta                    0;
    }
    un-named-reaction-1
    {
        type                irreversibleSurfaceArrheniusSurface;
        reaction              "NH3(s) + 3Ru(s) = N(s) + 3H(s)";
        A                    4.7625e+34;
        beta                  0;
        Ta                    12724.7;
    }
    ...
}

```

3.4. Converting units from cgs to SI

In the cgs system the rate-of-progress always has the unit of mol/cm^2s . The concentration of gas-phase species $[X_{k,G}]$ has the unit of mol/cm^3 while the concentration of site species $[X_{k,S}]$ has the unit of mol/cm^2 (assuming there is no bulk species in reactions). Then, we can balance the unit of left-hand-side and right-hand-side (only consider the first term) of the equation 16 as:

$$\frac{mol}{cm^2s} = \frac{1}{s} \left[\frac{mol}{cm^2} \right]^{1-\sum_i^S v'_{ki}} \left[\frac{mol}{cm^3} \right]^{-\sum_i^G v'_{ki}} \left[\frac{mol}{cm^2} \right]^{\sum_i^S v'_{ki}} \left[\frac{mol}{cm^3} \right]^{\sum_i^G v'_{ki}} \quad (18)$$

where G and S are number of gas-phase species reactants and site species reactants in the left-hand-side of the reaction.

In the SI system the rate-of-progress always has the unit of $kmol/m^2s$, and the concentration of a gas-phase species $[X_{k,G}]$ has the unit of $kmol/m^3$ while the concentration of a

site species $[X_{k,S}]$ has the unit of $kmol/m^2$ (assuming there is no bulk species in reactions). Then, we can balance the unit of left-hand-side and right-hand-side (only consider the first term) of the equation 16 as:

$$\frac{kmol}{m^2 s} = \frac{1}{s} \left[\frac{kmol}{m^2} \right]^{1-\sum_i^S v'_{ki}} \left[\frac{kmol}{m^3} \right]^{-\sum_i^G v'_{ki}} \left[\frac{kmol}{m^2} \right]^{\sum_i^S v'_{ki}} \left[\frac{kmol}{m^3} \right]^{\sum_i^G v'_{ki}} \quad (19)$$

Now, rewriting the unit of the rate constant (i.e., the first and second terms on the right-hand-side of the equation 19) in terms of cgs units yields:

$$\frac{1}{s} \left[\frac{kmol}{m^2} \right]^{1-\sum_i^S v'_{ki}} \left[\frac{kmol}{m^3} \right]^{-\sum_i^G v'_{ki}} = \frac{1}{s} \left[\frac{mol}{cm^2} \frac{10^3}{10^4} \right]^{1-\sum_i^S v'_{ki}} \left[\frac{mol}{cm^3} \frac{10^3}{10^6} \right]^{-\sum_i^G v'_{ki}} \quad (20)$$

or

$$\frac{1}{s} \left[\frac{kmol}{m^2} \right]^{1-\sum_i^S v'_{ki}} \left[\frac{kmol}{m^3} \right]^{-\sum_i^G v'_{ki}} = \frac{1}{s} \left[\frac{mol}{cm^2} \right]^{1-\sum_i^S v'_{ki}} \left[\frac{mol}{cm^3} \right]^{-\sum_i^G v'_{ki}} [10^{-1}]^{1-\sum_i^S v'_{ki}} [10^{-3}]^{-\sum_i^G v'_{ki}} \quad (21)$$

In other words, to convert from cgs unit to SI unit the rate constant in cgs system need to be multiplied by a coefficient, $coeff_s$, as:

$$coeff_s = \frac{1}{[10^{-1}]^{(1-\sum_i^S v'_{ki})} [10^{-3}]^{-\sum_i^G v'_{ki}}} = [10^{-1}]^{(\sum_i^S v'_{ki}-1)} [10^{-3}]^{\sum_i^G v'_{ki}} \quad (22)$$

Finally, the unit of the pre-exponential factor A_i in the rate constant must be multiplied by $coeff_s$ when converting units from cgs to SI.

As the same as in the gas-phase reaction, for activation energy E_i the conversion coefficient depends on the keyword of reaction line in the CHEMKIN file.

If there is no keyword (by default), E_i is in cal/mol . Then T_a in OF is calculated as:

$$T_a = \frac{E_i}{R_c} \quad (23)$$

where $R_c = 1.987316 \text{ cal/g.mol.K}$

If there is $JOULE/MOLE$, E_i is in J/mol . Then T_a in OF is calculated as:

$$T_a = \frac{E_i}{R} \quad (24)$$

where $R = 8.31451 \text{ J/kg.mol.K}$

3.5. Examples

3.6. Coverage modification rate constant reactions

In some cases there are experimental data that indicate that the Arrhenius expression for the rate constant in equation 17 is modified by coverage (concentration) of some surface or bulk species. Through use of auxiliary keyword *COV*, the rate constant for the forward reaction is modified as:

$$k_{fi} = A_i T^{\beta_i} \exp\left(-\frac{E_i}{RT}\right) \prod_{k=K_s^f(N_s^f)}^{K_s^l(N_s^l)} 10^{\eta_{ki}[Z_k(n)]} [Z_k(n)]^{\mu_{ki}} \exp\left(-\frac{\varepsilon_{ki}[Z_k(n)]}{RT}\right) \prod_{k=K_b^f(N_b^f)}^{K_b^l(N_b^l)} 10^{\eta_{ki}[a_k(n)]} [a_k(n)]^{\mu_{ki}} \exp\left(-\frac{\varepsilon_{ki}[a_k(n)]}{RT}\right) \quad (25)$$

where three coverage parameters are η_{ki} , μ_{ki} , and ε_{ki} for species k and reaction i th. η_{ki} and μ_{ki} are unitless. The unit of ε_{ki} is as same as E_i of other surface reactions.

The product in equation 25 runs over only those surface species that are specified as contributing to the coverage modification. Note that, the surface site fractions $[Z_k]$ appear in equation 25 rather than molar concentrations $[X_k]$ (mol/cm^2) for surface species, while bulk activities $[a_k]$ appear for bulk species. The term associated with μ_{ki} now makes it possible for rate-of-progress of a reaction to be proportional to any arbitrary power of a surface species concentration.

In CHEMKIN format:

```
// SURFACE MECHANISM OF WATER-GAS SHIFT REACTION OVER RHODIUM
//      H2/CO/O2/CO2/H2O OVER RH - SURFACE MECHANISM
//      References:
//      C. Karakaya, R. OtterstOtter, L. Maier, O. Deutschmann,
//      Applied Catalysis A: General, submitted (2013)
//      www.detchem.com/mechanisms

SITE/Rh_surface/      SDEN/2.72E-9/
  Rh(s)
  H2O(s)
  H(s)
  OH(s)
  CO(s)
  O(s)
  CO2(s)
  COOH(s)
END
REACTIONS      MWOFF      KJOULES/MOLE
```

H2 + Rh(s) + Rh(s) => H(s) + H(s)	3.000E-02	0.000	0.000
STICK			
...			
H2O(s) => Rh(s) + H2O	6.858E+14	-0.280	44.993
CO(s) => Rh(s) + CO	1.300E+13	0.295	
134.070			
COV/CO(s)	0.000E+00	0.000	-47.000/
CO2(s) => Rh(s) + CO2	3.920E+11	0.315	20.505
...			
END			

When converting into OF format,

```

reactions
{
  un-named-reaction-0 //
  {
    type      irreversibleSurfaceStickArrheniusSurface;
    reaction   "H2 + 2Rh(s) = 2H(s)";
    A          3.0e-2;
    beta       0;
    Ta         0;
  }
  ...
  un-named-reaction-7
  {
    type      irreversibleSurfaceArrheniusSurface;
    reaction   "H2O(s) = Rh(s) + H2O";
    A          4.747E+12;
    beta       -0.280;
    Ta         5411.383232;
  }
  un-named-reaction-8
  {
    type      irreversibleSurfaceCoverageModificationArrheniusSurface;
    reaction   "CO(s) = Rh(s) + CO";
    A          4.747E+12;
    beta       0.295;
    Ta         16124.82275;
    8
    (
      (Rh(s) (0 0 0))
      (H2O(s) (0 0 0))
      (H(s) (0 0 0))
      (OH(s) (0 0 0))
      (CO(s) (0 0 -5652.768473))
      (O(s) (0 0 0))
      (CO2(s) (0 0 0))
      (COOH(s) (0 0 0))
    );
  }
  un-named-reaction-9
  {
    type      irreversibleSurfaceArrheniusSurface;
    reaction   "CO2(s) = Rh(s) + CO2";
    A          4.747E+12;
    beta       0.315;
    Ta         2466.170586;
  }
}

```

} ...

3.7. Sticking coefficient reactions

For some simple surface reaction mechanisms, we have found it convenient to specify the surface reaction rate constant in terms of a sticking coefficient (probability), rather than an actual reaction rate. This approach is only allowed when there is exactly one gas-phase species reacting with a surface. Sticking-coefficient reactions may include any number of surface site or bulk-phase species as reactants, and any number of species of any phase as products.

In such cases, one might have a measurement or intuition about the probability that a certain process takes place when a collision between a given gas-species occurs with the surface. For consistency in expressing each surface reaction in terms of rate constant, we provide a conversion between this sticking coefficient form and the usual rate expression. The actual reaction rate (in *mole/cm²/sec*) is derived from this probability together with the physical gas-surface collision frequencies, as discussed further below.

The unitless sticking coefficients functional form has an Arrhenius-like form as follows:

$$\gamma_i = \min \left[1, a_i T^{b_i} \exp \left(-\frac{c_i}{RT} \right) \right] \quad (26)$$

In this case, γ_i , a_i and b_i are unitless and c_i has units compatible with RT , the gas constant used for reaction activation energies multiplied by temperature.

Conversion of a sticking coefficients γ_i to usual mass-action kinetic rate constants uses the collision frequency of the gas species with the solid surface, as shown:

$$k_i = \gamma_i \frac{\prod_{i=1}^S \sigma_i^{v'_{ij}}}{(\Gamma_{tot})^m} \sqrt{\frac{RT}{2\pi W_k}} \quad (27)$$

Here, R is the universal gas constant, W_k is the molecular weight of the gas-phase species. Γ_{tot} is the total surface site concentration summed over all surface phases (number of moles of surface sites per unit area), and m is the sum of all the stoichiometric coefficients of reactants that are surface species.

The term involving Γ_{tot} raised to the m power is needed to convert from the unitless sticking coefficient form to units appropriate for a rate constant, and the term in the square root accounts for the gas/surface collision frequency.

For example, in CHEMKIN format:

```
//      H2/O2 OVER RH - SURFACE MECHANISM
//      References:
//      C. Karakaya, O. Deutschmann, Chemical Engineering Science,
//      89 (2012) 171-184
//      www.detchem.com/mechanisms
SITE/Rh_surface/      SDEN/2.72E-9/
  Rh(s)
  H2O(s)
  H(s)
  OH(s)
  O(s)
END
REACTIONS      MWOFF      KJOULES/MOLE
H2 + Rh(s) + Rh(s) => H(s) + H(s)      3.000E-02      0.000      0.000
  STICK
O2 + Rh(s) + Rh(s) => O(s) + O(s)      1.000E-02      0.000      0.000
  STICK
H2O + Rh(s) => H2O(s)      1.000E-01      0.000      0.000
  STICK
H(s) + H(s) => Rh(s) + Rh(s) + H2      5.574E+19      0.239      59.691
O(s) + O(s) => Rh(s) + Rh(s) + O2      5.329E+22      -0.137      386.995
H2O(s) => Rh(s) + H2O      6.858E+14      -0.280      44.993
O(s) + H(s) => OH(s) + Rh(s)      8.826E+21      -0.048      73.365
OH(s) + Rh(s) => O(s) + H(s)      1.000E+21      0.045      48.041
OH(s) + H(s) => H2O(s) + Rh(s)      1.743E+22      -0.127      41.731
H2O(s) + Rh(s) => OH(s) + H(s)      5.408E+22      0.129      98.220
OH(s) + OH(s) => O(s) + H2O(s)      5.736E+20      -0.081      121.594
O(s) + H2O(s) => OH(s) + OH(s)      1.570E+22      0.081      203.407
END
```

When converting into OF format,

```
reactions
{
  un-named-reaction-0
  {
    type      irreversibleSurfaceStickArrheniusSurface;
    reaction   "H2 + 2Rh(s) = 2H(s)";
    A          3.0e-2;
    beta       0;
    Ta         0;
  }
  un-named-reaction-1
  {
    type      irreversibleSurfaceStickArrheniusSurface;
    reaction   "O2 + 2Rh(s) = 2O(s)";
    A          1.0e-2;
    beta       0;
    Ta         0;
  }
  ...
}
```

```
}
```

References

- [1] R. J. Kee, F. M. Rupley, E. Meeks, J. A. Miller, CHEMKIN-III: A FORTRAN chemical kinetics package for the analysis of gas-phase chemical and plasma kinetics, Tech. Rep. SAND96-8216, Sandia National Labs., Livermore, CA, USA (1996).